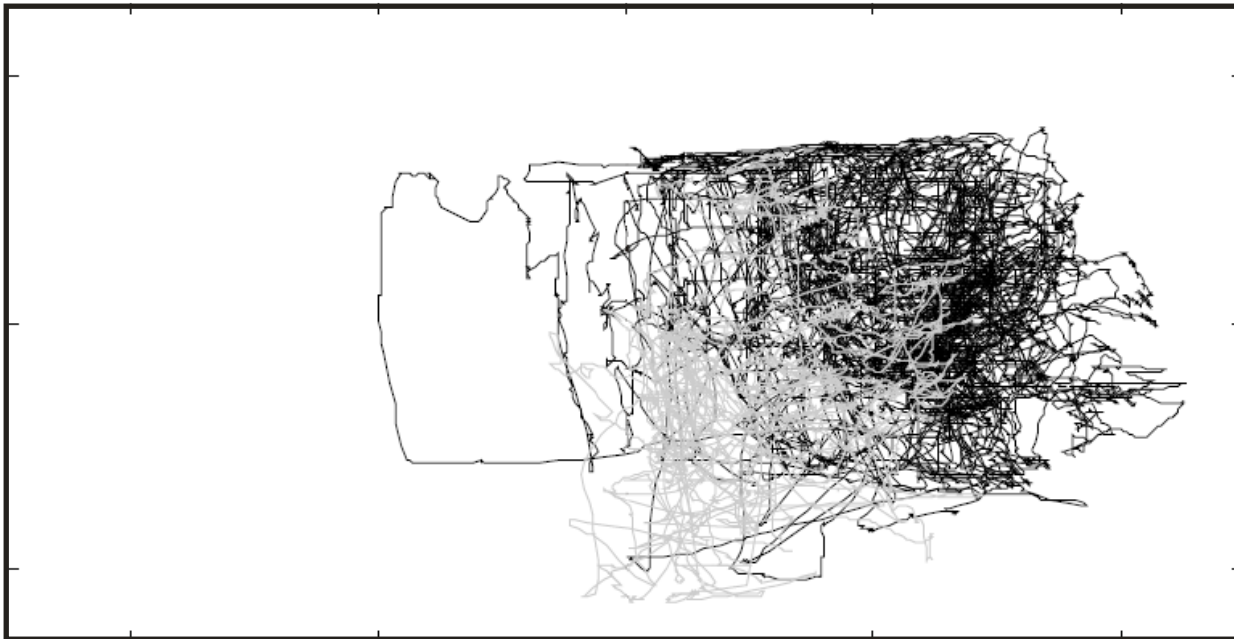


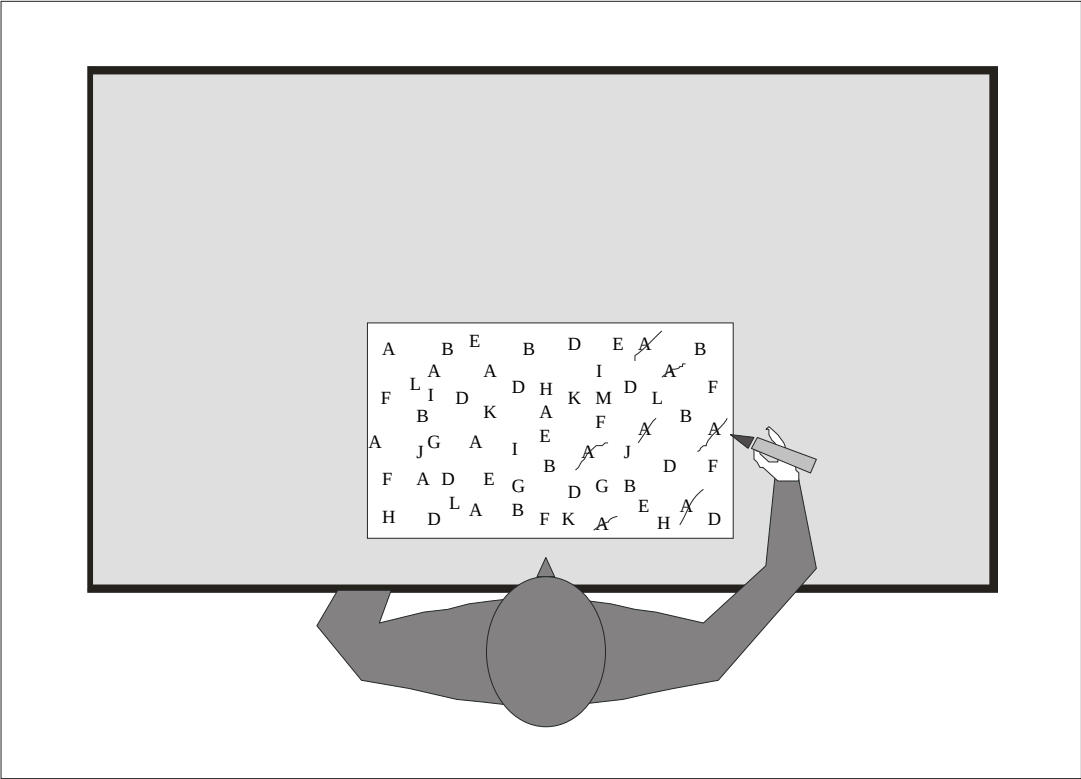
Neglect

Complete darkness

Patient group with neglect (N=12)



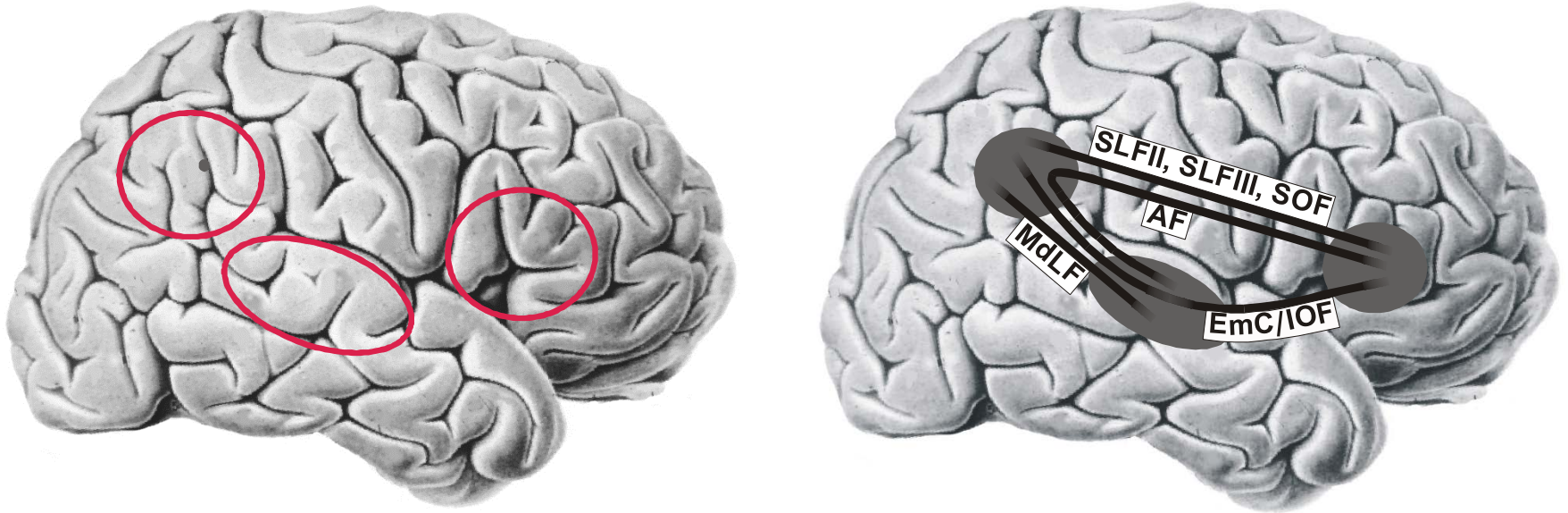
- Waiting / Relaxing / Doing nothing
- Active search



Bisiach & Luzzatti (Cortex 1978)

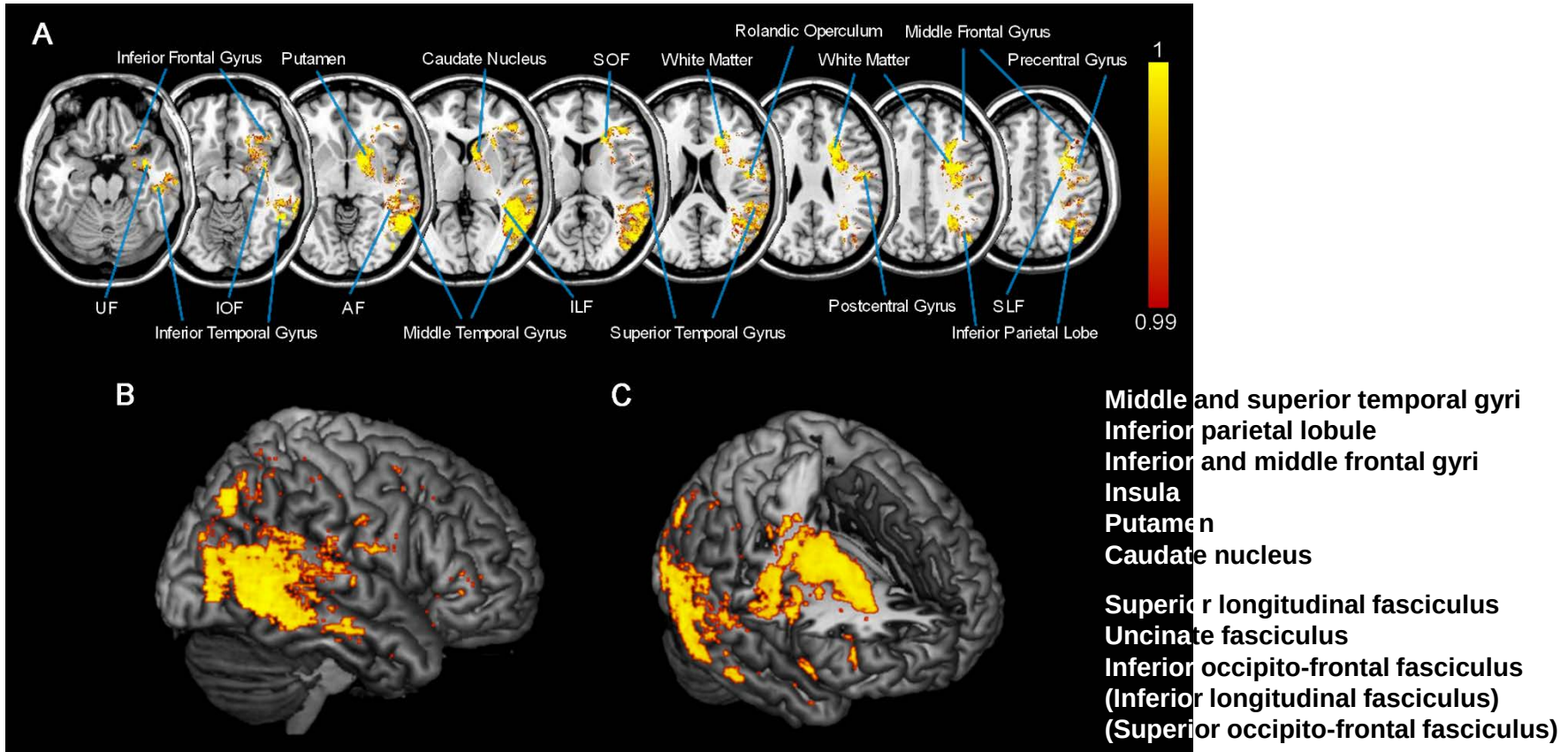


Anatomy



Perisylvian network for spatial attention and orientation

Anatomy

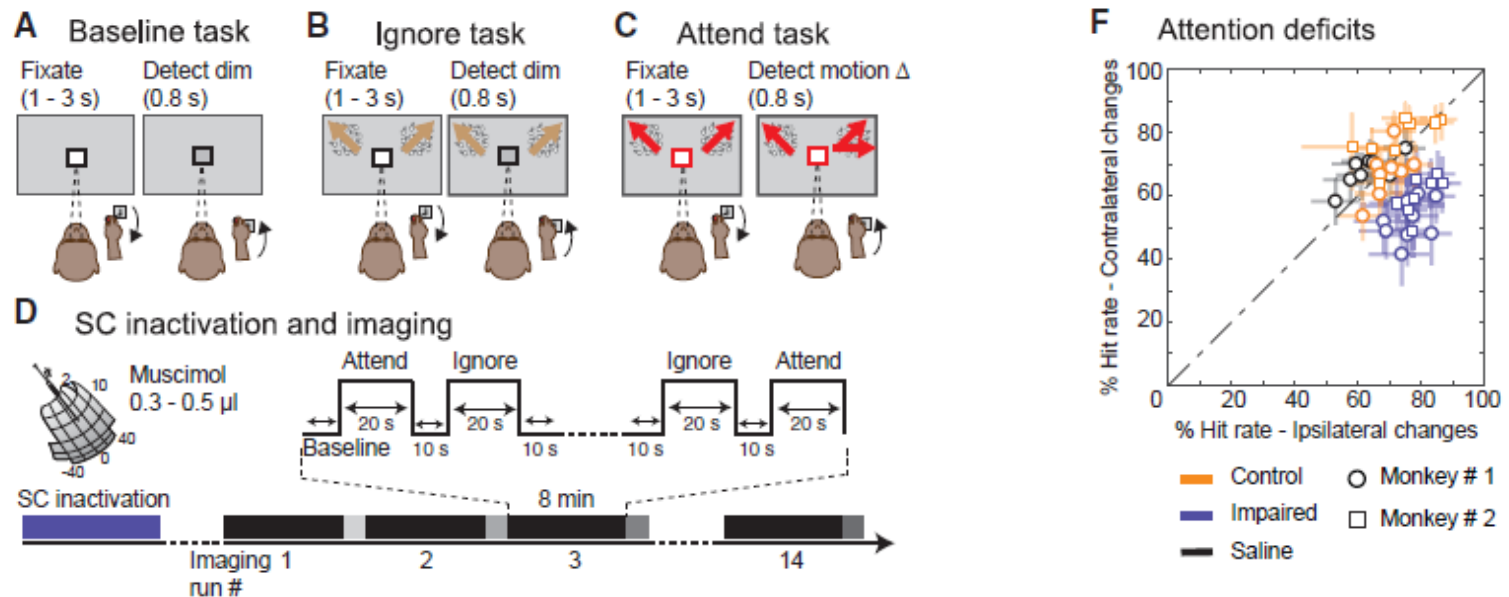


Perisylvian network including parietal, temporal, and frontal grey matter regions as well as interconnecting WM fibres

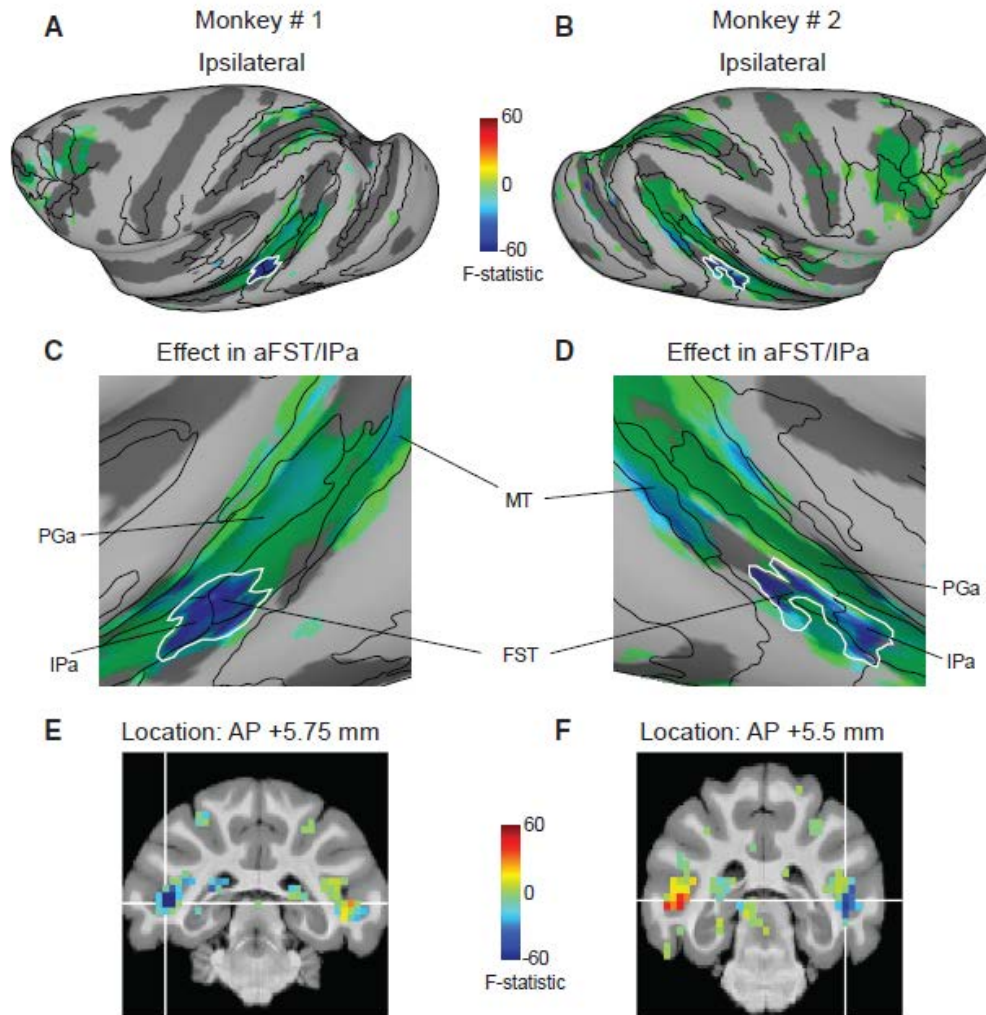
Spatial Attention Deficits Are Causally Linked to an Area in Macaque Temporal Cortex

Amarendra R. Bogadhi,^{1,*} Anil Bollimunta,¹ David A. Leopold,^{2,3} and Richard J. Krauzlis^{1,4,*}

Behavior and fMRI activity across the cerebral cortex during experimental induction of neglect through reversible inactivation of the superior colliculus



During inactivation, monkeys exhibited attentional deficits of neglect in tasks using either motion or non-motion stimuli.



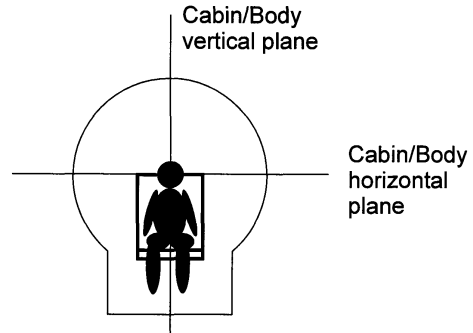
Behavioral neglect was accompanied by marked reductions in fMRI modulation that was strongest in a small region on the floor of the superior temporal sulcus

Spatial neglect

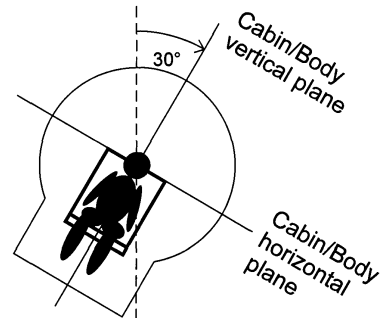
The characteristic deficit of patients with neglect is a deviation of exploratory movements to the ipsilesional (right) side of space.

The centre of exploration does no longer correspond with the sagittal body mid-plane; the center is shifted towards the right side.

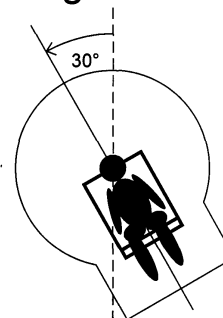
Upright



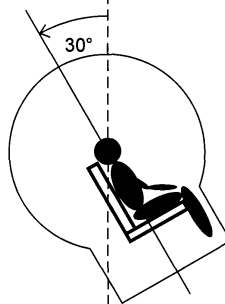
Leftward



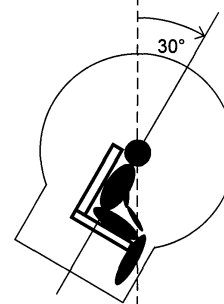
Rightward



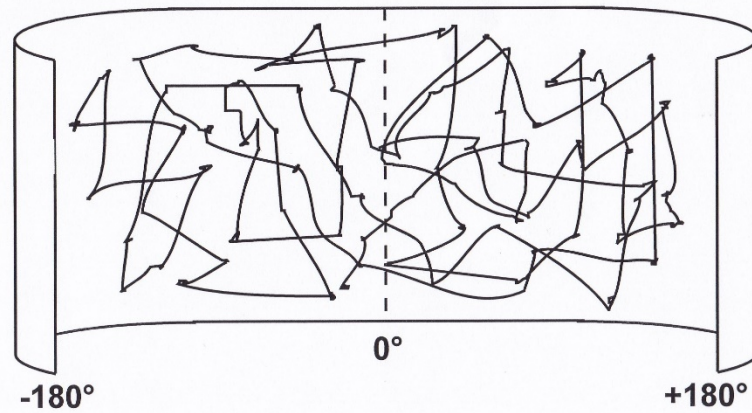
Backward



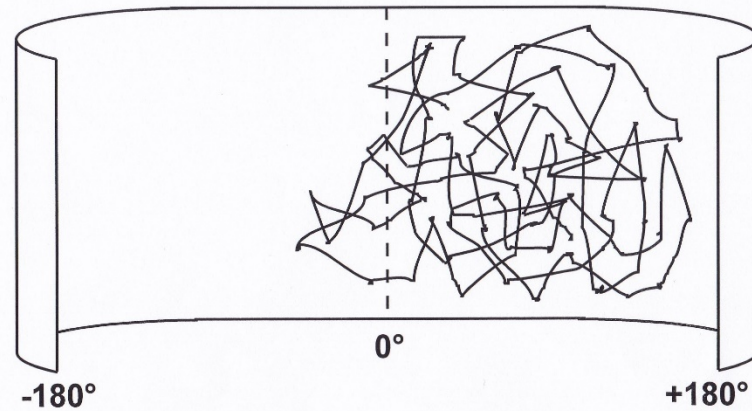
Forward



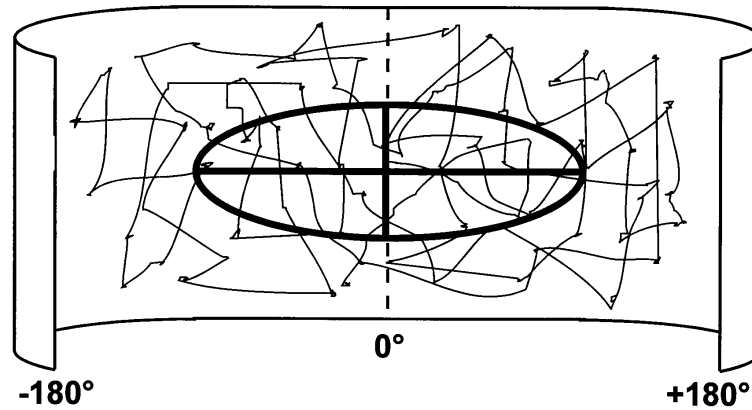
Normal
subject



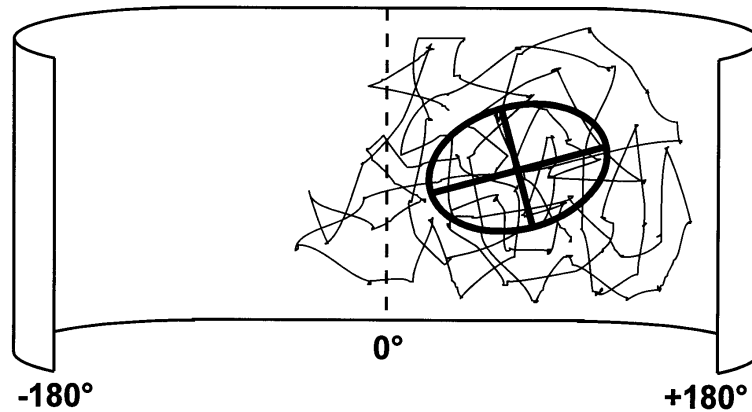
Neglect
patient



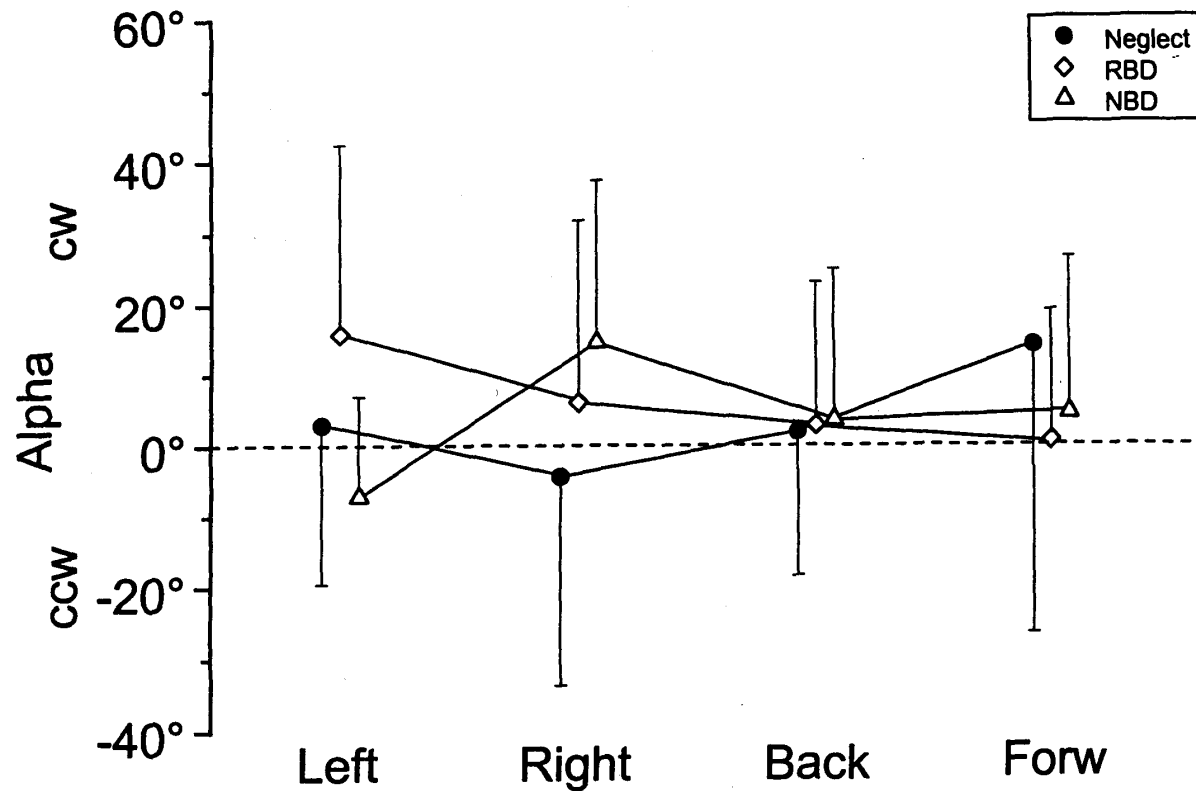
Normal
subject



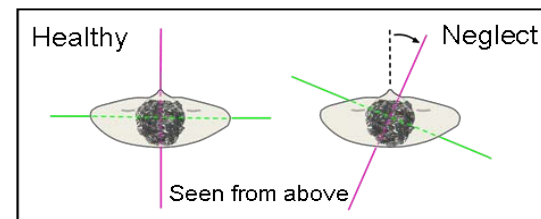
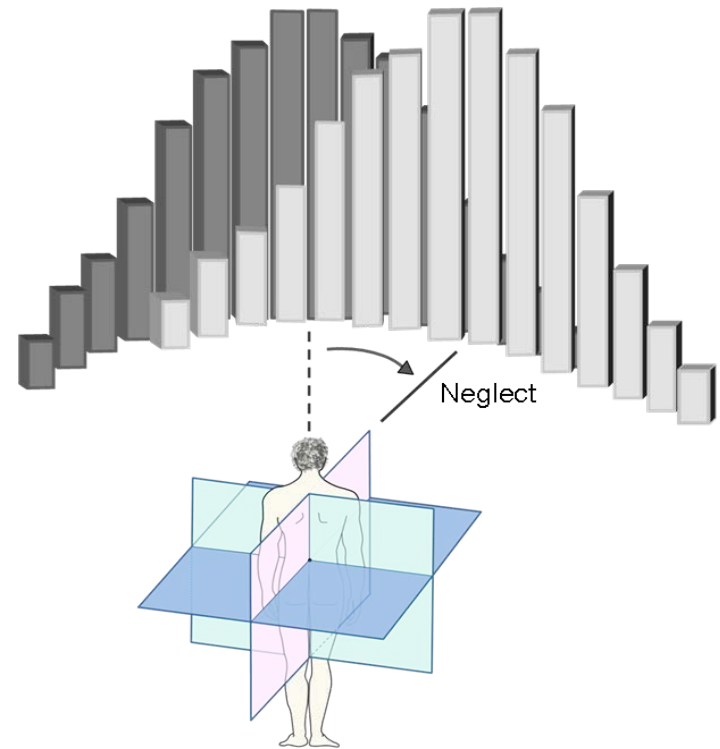
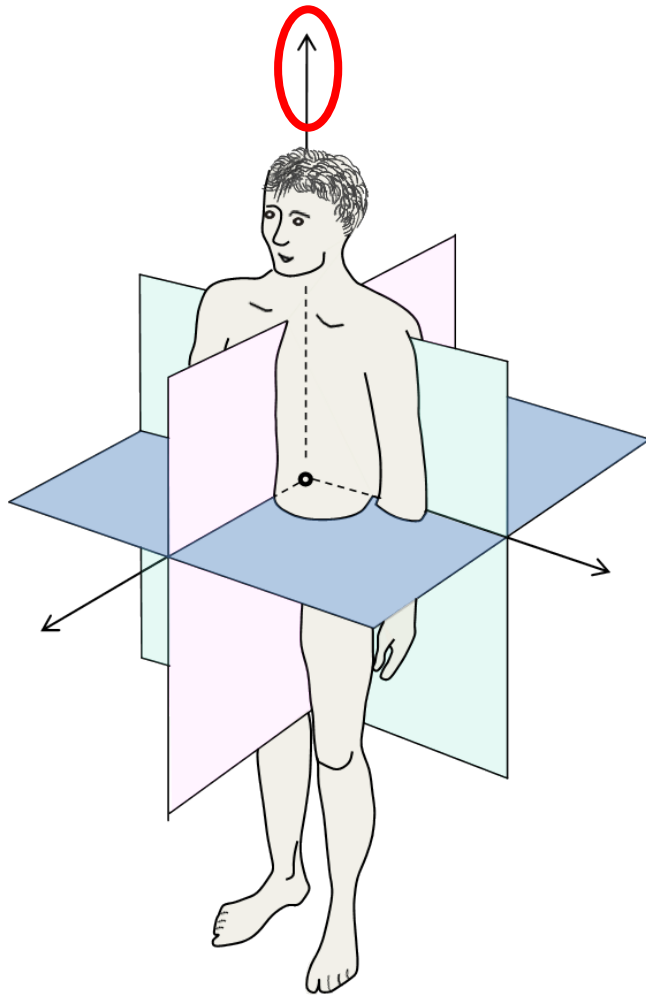
Neglect
patient



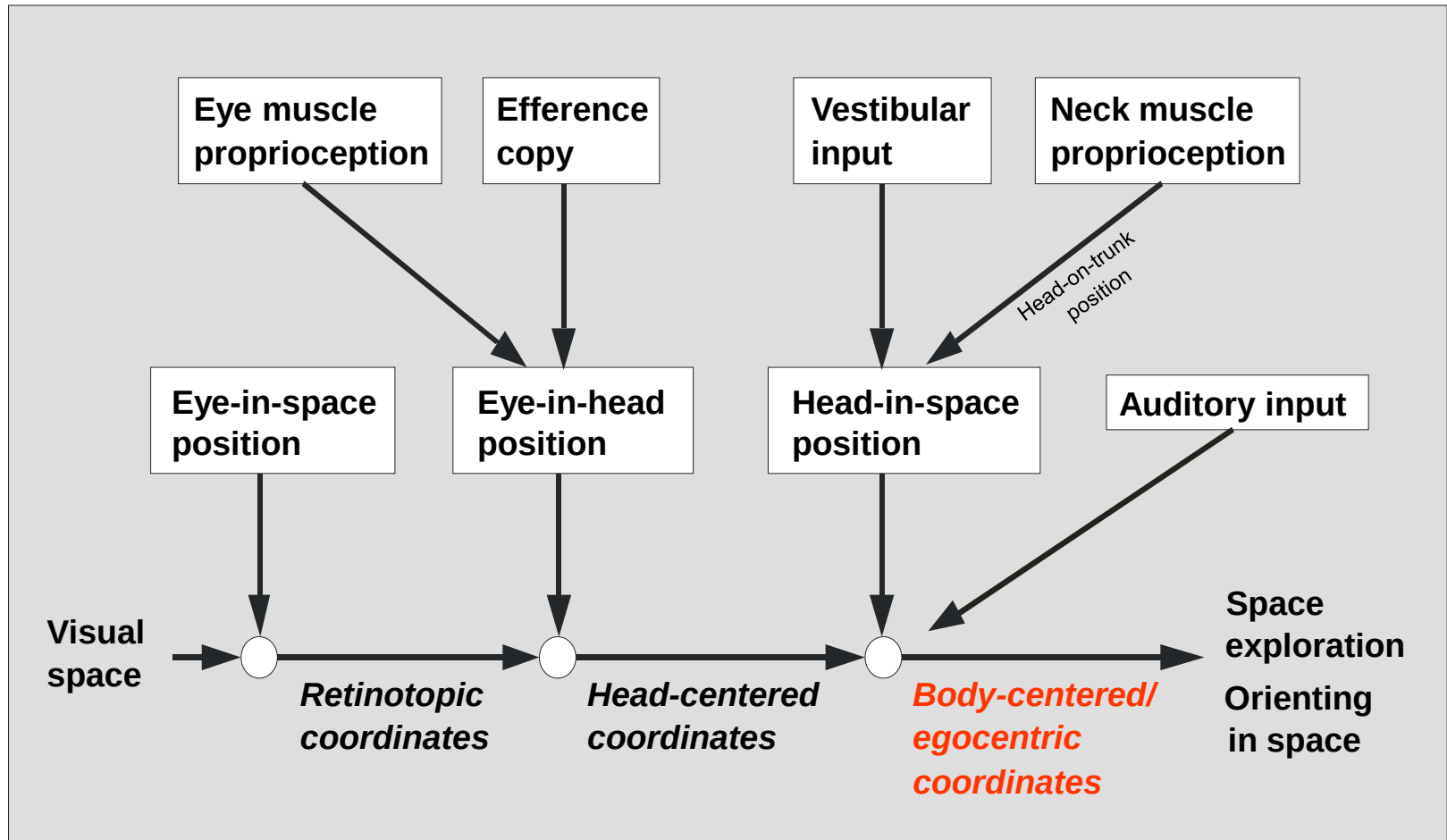
No indication for gravity-based reference of spatial neglect



Rotation of the egocentric spatial frame of reference towards the ipsilesional right

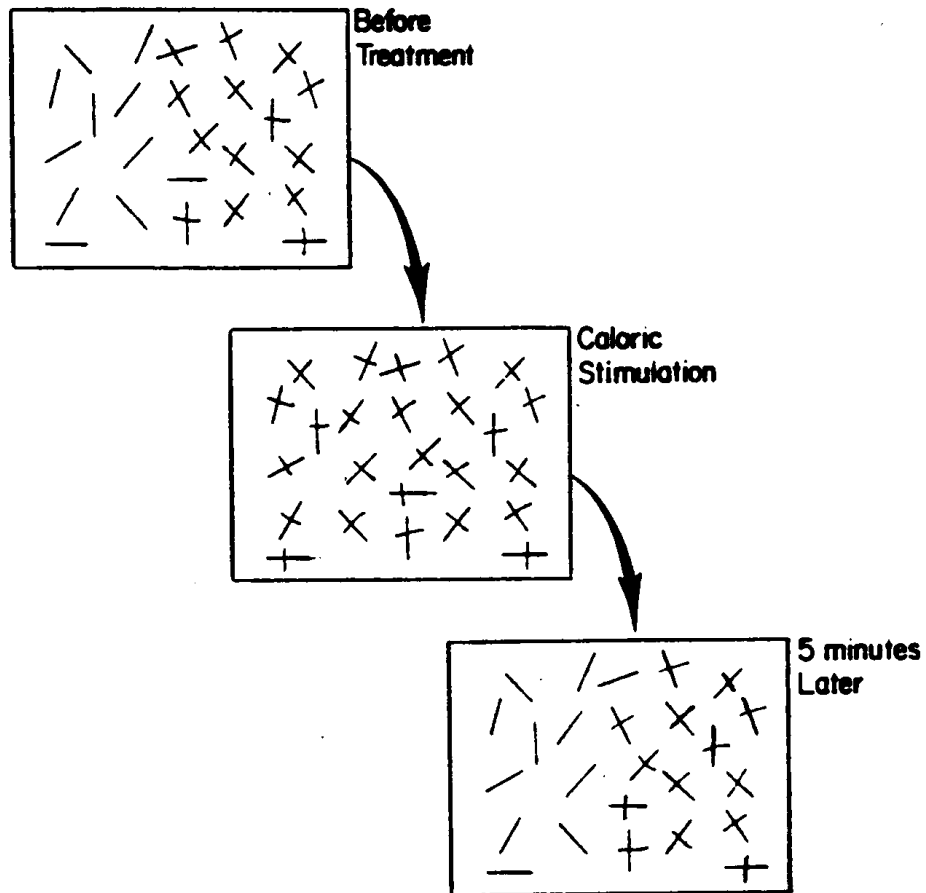


Transformation hypothesis



(Karnath, Neuropsychol Rehabil 1994)

Caloric Vestibular Stimulation



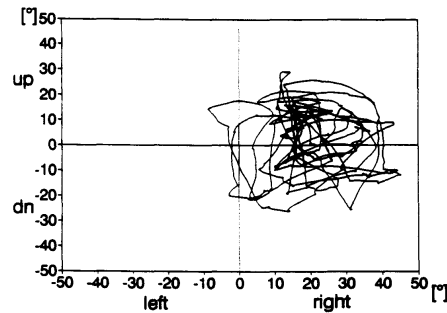
Neck muscle proprioception

Vibration

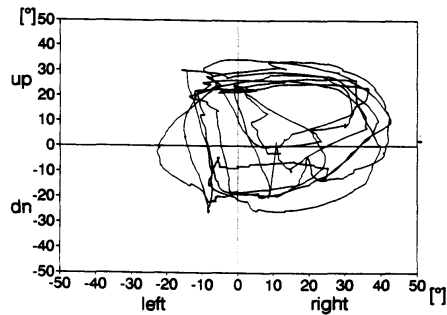


Neglect patient

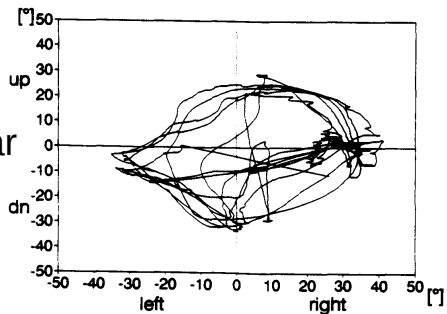
Spontaneous
visual search



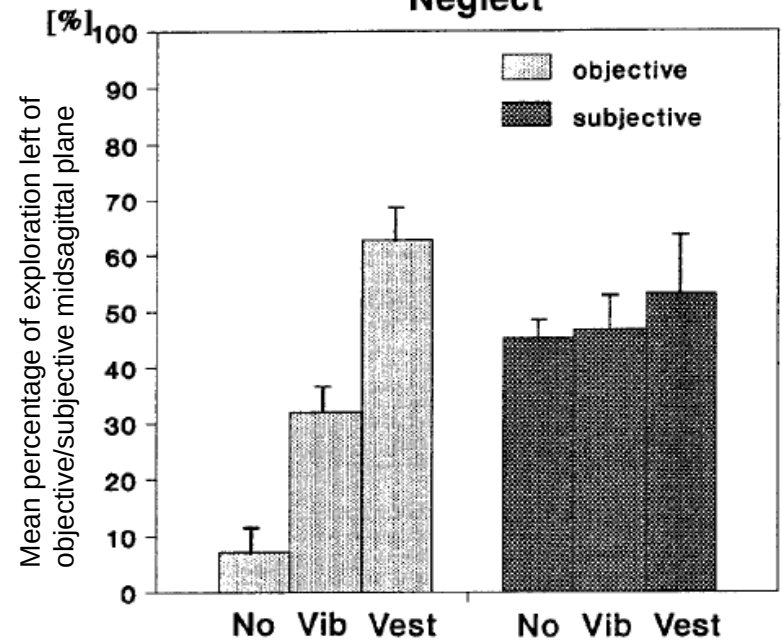
Visual search
under neck
vibration



Visual search
under vestibular
stimulation

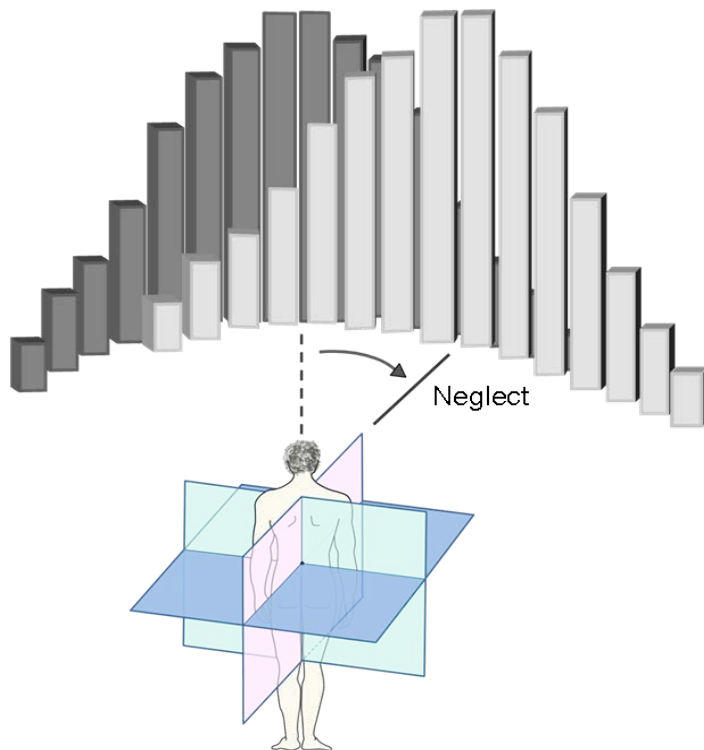


Neglect



Conclusions

The biased distribution of exploratory activity results from an altered representation of own body position with respect to external objects.



In neglect patients, this body-centered matrix (or egocentric reference frame of topographical information) is rotated towards a new 'default position' on the right.

